Master Architecture & Baseline Integration

System Architecture Specification: Epistemic Models, Lineage Semantics, and Verification Topology

Status: Baseline Working Draft | **Version:** 1.0.0 | **Classification:** Technical Architecture

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Executive Summary: Architectural Closure

The APEDP platform specification currently details strategic justification, technical requirements, scientific assurance, and discovery metrology across four core documents. The primary remaining gap is not conceptual, but architectural: **integration semantics, unified lifecycle governance, and formal operational wiring**. This master baseline integrates these views into an authoritative specification, binding data pipelines, symbolic models, simulation fabrics, and laboratory execution planes into a buildable, verified system.

Key Architectural Objective: *Replace document proliferation with an authoritative Master Baseline and Requirements Traceability Matrix (RTM). Ensure every epistemic claim is traceable to invariant physical foundations and verified instruments.*

1. Master System Integration Architecture (P0)

The platform topology connects seven foundational planes into a continuous feedback loop:

Topology Chain: Data Plane ↔ Knowledge Engine ↔ Physics Engine ↔ Simulation Fabric ↔ AI Agent Orchestration ↔ Empirical Lab Execution ↔ Engineering Core

• **Service Boundaries & APIs:** High-throughput gRPC interfaces govern command and control pipelines, while zero-copy Apache Arrow Flight buffers handle large-scale tensor and dataset transfers.
• **Event & Data Contracts:** Strict contract-first schemas (Avro/JSON Schema) enforce forward and backward compatibility across asynchronous streaming boundaries.
• **Authority & Security Zones:** RBAC and ABAC isolate physical experiment actuators, high-performance simulation clusters, and agent exploration environments into distinct security zones.
• **Dependency Orchestration:** DAG-based workflows prevent hidden cyclical dependencies; simulation-experiment feedback is modeled explicitly as controlled iterative loops.

2. Formal Epistemic Type System & Lineage Semantics

To prevent epistemic corruption (such as conflating approximations with empirical observations), the platform defines machine-readable types across all entities:

Entity Type	Definition	Verification Standard	Lifecycle
Observation	Raw empirical sensor capture	Cryptographic hardware signature	Immutable
Measurand	Quantity intended to be measured	Standard SI / Metrological definition	Static
State	Parameterization of observables at t■	Coordinate & conservation audit	Versioned
Equation	Mathematical invariance relation	Dimensional & gauge proof	Immutable
Model	Parameterized numerical implementation	Domain of validity & bounds check	Versioned
Hypothesis	Testable, falsifiable proposition	Explicit falsification criteria	Ephemeral
Prediction	Numerical assertion from model	A posteriori error bounds (±ε)	Immutable
Experiment	Executable falsification protocol	Standardized laboratory runbook	Immutable

Evidence	Distance: Prediction vs. Observation	Multi-party reproduction consensus	Ledgered
Claim	Synthesis from verified Evidence	Adversarial peer-challenge passage	Ledgered
Decision	Resource allocation or promotion	Cryptographic signature (Human/Agent)	Audited

Lineage Edge Semantics: Edges in the knowledge graph explicitly define causal and mathematical relationships: DERIVED_FROM (deterministic math), INFERRED_BY (statistical induction), ASSUMED_BY (unverified axiom), MEASURED_BY (sensor run), SIMULATED_BY (solver run), APPROXIMATED_FROM (linearization/truncation), and CONTRADICTED_BY (empirical invalidation). Queries can isolate root unverified assumptions behind any claim instantly.

3. Full Lifecycle Management & Traceability Matrix

To eliminate 'zombie science'—deprecated models remaining silently active—all artifacts adhere to a strict state machine:

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Universal Promotion Cycle:   Draft ■■■ Candidate ■■■ Validated ■■■ Operational ■■■ Superseded ■■■ Retired
```

Requirements Traceability Chain: Goal → Requirement → Design → Implementation → Test → Evidence.
Every system requirement must declare: *Requirement_ID*, *Owner*, *Design_Reference (ADR)*, *Implementation_Artifact (Git commit/digest)*, *Acceptance_Test (script)*, and *Validation_Evidence*.

4. Operational Scaling & Simulation Infrastructure

• **Digital Twin of the Research Organization (DTRO):** Closed-loop modeling of compute clusters, energy draw, instrument schedules, lab consumables, and staffing queues to identify discovery bottlenecks prior to resource commitment.
• **Scientific Data Federation:** Distributed querying layer (DuckDB / Apache Arrow Flight) uniting cloud buckets, observatories, and edge lab instruments without centralized data duplication.
• **Synthetic-Universe Generator:** Automated benchmark worlds embedding known physical laws alongside hidden injected anomalies to rigorously measure agent discovery sensitivity, false discovery rates, and search efficiency.

5. Epistemic Rigor, Verification, & Deliberation

• **Conservation & Invariant Registry:** Central verification registry enforcing Noether invariants (energy, momentum, charge, entropy) before admitting models to production.
• **Dimensional Ontology & Nondimensionalization:** First-class unit representations halting compile execution on inhomogeneous operations (e.g., e^(-5 J)); automated Buckingham-π transformations to optimize numerical stability.
• **Numerical Error Certification:** Rigorous a posteriori error estimators distinguishing numerical discretization artifacts from genuine physical discrepancies.
• **Hardware Calibration Traceability:** Sensor → Calibration Standard → Calibration Procedure → Measurement.
• **Adversarial Deliberation Engine:** Multi-agent falsification tournament (Generate → Challenge → Counterexample → Refine → Falsify).
• **Blind Replication Packaging:** Self-contained containerized bundles allowing external validation without access to generative priors.

6. Socio-Technical Governance & Knowledge Preservation

Workflow Stage	Autonomous AI Role	Human Specialist Role	Escalation Gate
Hypothesis Generation	Combinatorial exploration & priors	Contextual intuition & goal framing	Automated bounds check
Dimensional & Invariant Audit	Rigorous formal verification	Review exceptional edge cases	Zero tolerance for violations
Simulation Execution	Dynamic solver selection & tuning	Resource threshold approvals	Compute & energy budgets
Physical Experimentation	Protocol synthesis & optimization	Final signoff & physical execution	Mandatory human approval
Claim Promotion	Evidence aggregation & report	Authoritative epistemic signoff	Multi-stakeholder signoff

• **Discovery Prioritization Vector:** S = [Novelty, Predictive Reach, Generality, Reproducibility, Engineering Leverage]. Evaluated via Pareto frontier optimization rather than a single collapsed scalar.
• **Scientific Memory Layer:** Long-horizon storage decoupling institutional memory from neural model weights using open, self-describing standards (OWL/RDF, Parquet, Zarr) and containerized execution receipts.